

5G Core Conformance & Performance Testing — Landscape

Standards, organizations, tools, and where upfbench fits

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1. The big picture

There is **no single “5G core” certification stamp** the way mobile devices are certified. Formal 3GPP **conformance + certification** programs target **devices (UEs)** and **base stations** — not core network functions. For the **5G core**, correctness is validated through three overlapping activities:

1. **Protocol conformance** — does each network function (NF) obey the 3GPP stage-3 protocol spec for its interface (e.g. N4/PFCP = TS 29.244, SBI = TS 29.500-series)?
2. **Interoperability** — does vendor A’s NF work with vendor B’s? (ETSI Plugtests, GSMA.)
3. **Load / performance / NFV** — capacity, throughput, latency under realistic subscriber load (ETSI NFV-TST, RFC 2544).

upfbench is an **open-source** tool that covers a slice of (1) and (3) for the **UPF** over the **N4 interface** — the same job commercial N4 emulators and core load testers do, using open components (**pfcp**sim + **tcp**replay) instead of paid systems.

2. Standards & specifications

Spec	Body	What it defines	Used for
TS 29.244	3GPP	PFCP — the SMF–UPF N4 control protocol (PDR/FAR/QER/URR, association, session est/mod/del)	upfbench Suite 3 asserts UPF compliance here
TS 29.500-series	3GPP	SBI — HTTP/2 REST APIs between NFs (Namf, Nsmf, Nudm, ...)	SBI conformance (separate area)
TS 23.501 / 23.502	3GPP	5G system architecture & procedures (stage 2)	Defines the behaviour being tested
TS 38.523-1	3GPP/ETSI	UE conformance test spec (5GS)	Device certification (RAN5)
TS 38.141	3GPP	Base station conformance	gNB testing
ETSI GS NFV-TST 009	ETSI NFV ISG	NFV data-plane performance methodology	upfbench Suite 1 methodology
RFC 2544 / RFC 8219	IETF	Throughput/latency benchmarking (NDR/PDR)	upfbench Suite 1 methodology

3. Organizations

Organization	Role in testing	Scope
3GPP (RAN5, CT4, SA5)	Writes the protocol & conformance specs	RAN5 = UE conformance; CT4 = core protocols (N4, SBI); SA5 = management
GCF (Global Certification Forum)	Mandates 3GPP test cases to certify devices	Devices / IoT — <i>not core NFs</i>
PTCRB	North-American device certification	Devices
GSMA	Interoperability testing & network requirements	VoLTE/5G/IMS interop, network slicing templates
ETSI TC INT	Core Network and Interoperability Testing	Core NF interop, test specs
ETSI Plugtests	Multi-vendor interoperability events	Hands-on interop validation
ETSI NFV ISG (TST WG)	NFV performance & test methodology	Virtualised NF performance
O-RAN Alliance / OTIC labs	Open RAN conformance & interop	RAN (adjacent to core)

4. Commercial test tools

Tool / vendor	What it provides	Relevant to
GL Communications — MAPS 5G N4 Emulator	Emulates SMF & UPF per TS 29.244; regression, edge-case, conformance & performance on N4	N4/PFCP conformance (same as our Suite 3)
Emblasoft (Evolver)	Simulates the SMF toward a UPF per TS 29.244; N4 functional + load	N4 conformance + load
Valid8	Per-NF conformance test suites (e.g. AMF, SMF)	SBI + N-interface conformance
Spirent Landslide	Emulates millions of subscribers, control + data plane; SA/NSA	Core load / performance (our Suites 1–2)
Keysight LoadCore	Emulates AMF/SMF/UPF; tens of thousands of events/s, hundreds of Gbps; node + E2E	Core load / performance ; validated 100 Gbps UPF (NEC)
Rohde & Schwarz	3GPP base-station conformance	RAN side
Vendor self-declarations (e.g. Cisco UPF “N4 3GPP Compliance” doc)	Per-release statement of which TS 29.244 IEs/procedures are supported	N4 conformance evidence

5. Open-source tools

Tool	What it provides
pfcpsim (omec-project)	Emulates an SMF over N4 (PFCP); drives association + session procedures — <i>the N4 driver upfbench uses</i>
TRex (Cisco)	High-rate traffic generator (DPDK) for UPF throughput testing
iperf / iperf3	Uplink/downlink peak-rate, TCP/UDP benchmarking

Tool	What it provides
tcpreplay	Replays crafted GTP-U pcaps on N3 — <i>the generator upfbench uses</i>
UERANSIM / gNBsim	Emulated gNB + UE for end-to-end 5G data-path testing
Open5GS / free5GC / OAI / SD-Core	Open-source 5G cores (the systems under test)

6. Where upfbench fits

upfbench suite	Equivalent commercial category	Standard	What it does
Suite 1 — performance	Spirent Landslide / Keysight LoadCore	RFC 2544, NFV-TST 009	Throughput (NDR/PDR), latency/jitter, burst, multi-flow on N3→UPF→N6
Suite 2 — load	Spirent Landslide / Keysight LoadCore	—	Session capacity, per-UE throughput, latency under N UEs
Suite 3 — pfcf conformance	GL MAPS / Emblasoft / Valid8	TS 29.244	N4 procedure conformance: association, session establish/modify/delete, unknown-SEID rejection

Honest scope vs. commercial conformance suites: upfbench Suite 3 covers the N4 **procedures** (happy-path + unknown-SEID negative). Commercial suites additionally do exhaustive **IE-level and malformed-message** validation with exact Cause-code checks (the “raw-PFCP harness” upfbench’s CF-05 notes as future work). upfbench’s value is being **open, reproducible, and UPF-agnostic** (one adapter per UPF), at near-zero cost.

7. Sources

- 3GPP RAN WG5 — <https://www.3gpp.org/3gpp-groups/radio-access-networks-ran/ran-wg5>
- 3GPP / GCF 5G device certification — <https://www.3gpp.org/news-events/partner-news/gcf-5g>
- ETSI TS 138 523-1 (5GS UE conformance) — https://www.etsi.org/deliver/etsi_ts/138500_138599/13852301/15.01.00_60/ts_13852301v150100p.pdf
- Keysight — 5G conformance testing process & coverage — <https://www.keysight.com/blogs/en/inds/2020/05/29/5g-testing-conformance-testing-process-and-coverage-matter>
- Rohde & Schwarz — 3GPP base-station conformance — https://www.rohde-schwarz.com/us/solutions/wireless-communications-testing/mobile-network-infrastructure-testing/3gpp-base-station-conformance-testing/3gpp-base-station-conformance-testing_256119.html
- GL Communications — MAPS 5G N4 Emulator — <https://www.gl.com/5G-N4-interface-emulator-using-maps.html>
- Emblasoft — N4 interface testing — <https://emblasoft.com/blog/comprehensive-testing-and-simulation-of-the-n4-interface-is-key-to-creating-multi-vendor-cloud-native-5g-networks>
- Cisco — N4 Interface 3GPP Compliance (PDF) — https://www.cisco.com/c/en/us/td/docs/wireless/ucc/upf/2021-01/b_ucc-5g-upf-config-and-admin-guide_2021-01/m_5g-upf-3gpp-dec-2018-n4-spec-compliance.pdf
- Valid8 — 5G AMF Conformance Test Suite — <https://www.valid8.com/datasheets/5g-amf-conformance-test-suite>
- PFCP (overview) — <https://en.wikipedia.org/wiki/PFCP>
- ETSI TC INT (Core Network & Interoperability Testing) — <https://www.etsi.org/committee/1401-int>
- GSMA Interoperability Testing — <https://www.gsma.com/services/interoperability-testing/>

- Spirent Landslide — <https://www.spirent.com/products/core-network-test-5g-lte-ims-wifi-diameter-landslide>
- Keysight LoadCore (datasheet PDF) — <https://www.keysight.com/us/en/assets/3120-1180/data-sheets/LoadCore-5G-Core-Testing.pdf>
- Open-source tools to benchmark 5G core UPF (Medium) — https://medium.com/@googler_ram/open-source-tools-to-benchmark-5g-core-upf-bfc46f05eb78

Compiled June 2026. Web sources accessed via search; specification section numbers per 3GPP Release 16/17.